

Passive Cell Balancing and Thermal Design

Battery Management System for Electric Scooter | ECE 445 Team 35 | Technical Design Note

Design Objective

The slave board uses the LTC6811 passive balancing outputs to selectively discharge cells with higher voltage. The balancing path must remove charge from an enabled cell while keeping sense-node disturbance and resistor heating within acceptable limits.

- Prototype acceptance target: balancing current ≥ 20 mA per enabled cell at 4.2 V.
- Original sizing target: correct an 8%-10% SOC imbalance in approximately 1 hour for a 5 Ah pack.
- Main thermal concern: localized heating in the bleed resistor and nearby MOSFET switching devices.

Balancing Current Derivation

The required balancing current is estimated from the charge that must be removed from the high-SOC cell divided by the balancing time.

$$I_{\text{bal}} = (\text{SOC_imbalance} * Q_{\text{nom}}) / t_{\text{balance}}$$

$$I_{\text{bal}} = (0.10 * 5 \text{ Ah}) / 1 \text{ h} = 0.5 \text{ A}$$

This 0.5 A value represents the aggressive analytical target for rapidly correcting a 10% imbalance in a 5 Ah pack. The prototype verification target was lower, requiring at least 20 mA at 4.2 V.

Parameter	Value	Meaning
SOC_imbalance	0.08 to 0.10	Expected cell imbalance range considered in the design analysis.
Q_nom	5 Ah	Nominal capacity used for initial balancing-current sizing.
t_balance	1 h	Desired correction time for the initial sizing calculation.
I_bal	0.5 A	Current required to correct 10% imbalance in 1 hour for 5 Ah.

Bleed Resistor Selection

For passive balancing, the discharge resistor is selected using Ohm's law at the maximum cell voltage. A lower resistor increases balancing current but also increases heat dissipation.

$$R_{\text{bal}} = V_{\text{cell}} / I_{\text{bal}}$$

$$R_{\text{bal}} = 4.2 \text{ V} / 0.5 \text{ A} = 8.4 \text{ Ohm}$$

A 10 Ohm resistor was selected as the nearest practical value. This slightly reduces current compared with the 8.4 Ohm ideal calculation, but it simplifies component selection and provides a more conservative thermal condition.

$$I_{\text{bal},10\text{ohm}} = V_{\text{cell}} / R = 4.2 \text{ V} / 10 \text{ Ohm} = 0.42 \text{ A}$$

Power Dissipation and Thermal Risk

The bleed resistor converts removed cell energy directly into heat. At high cell voltage, the resistor power is calculated from the squared voltage divided by resistance.

$$P_{\text{bleed}} = V_{\text{cell}}^2 / R_{\text{bal}}$$

$$P_{\text{bleed}} = (4.2 \text{ V})^2 / 10 \text{ Ohm} = 1.76 \text{ W}$$

Because 1.76 W is a significant local heat source on a compact PCB, resistor power rating alone is not sufficient. Heat must also be spread through copper area and thermal vias so nearby MOSFETs and sense traces remain inside safe operating limits.

Design Issue	Technical Impact	Mitigation
Resistor self-heating	Can raise local PCB temperature during extended balancing.	Use 3 W resistor rating and large copper area.
MOSFET proximity	Nearby switching MOSFETs can be thermally stressed.	Move heat away with vias and careful placement.
Sense-node disturbance	Balancing current can perturb cell voltage readings.	Route discharge paths away from precision sense nodes and sample after settling when needed.

Verified Prototype Behavior

The prototype requirement was verified by enabling balancing at 4.2 V and measuring the discharge current with a digital multimeter.

Measured $I_{bal} = 22.4 \text{ mA}$ at $V_{cell} = 4.2 \text{ V}$

This satisfies the $\geq 20 \text{ mA}$ per-cell acceptance criterion. The measured prototype current is appropriate for demonstrating controlled passive balancing, while the analytical resistor sizing documents the higher-current design considerations and thermal constraints.